

A Structure-Preserving Reduced-Order Model for Particle-in-Cell Simulations

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Landau Damping

Landau damping is a phenomena where the energy of the electric field is damped by wave-particle interactions in a collisionless plasma. This is a classic test problem for particle-in-cell (PIC) codes where the quantity of interest is the damping rate. Simulated in 2D2V with the initial condition

$$f(t=0, x, y, v_x, v_y) = \frac{1}{2\pi} \exp\left(-\frac{v_x^2 + v_y^2}{2}\right) [1 + \alpha \cos(kx)] [1 + \alpha \cos(ky)]$$

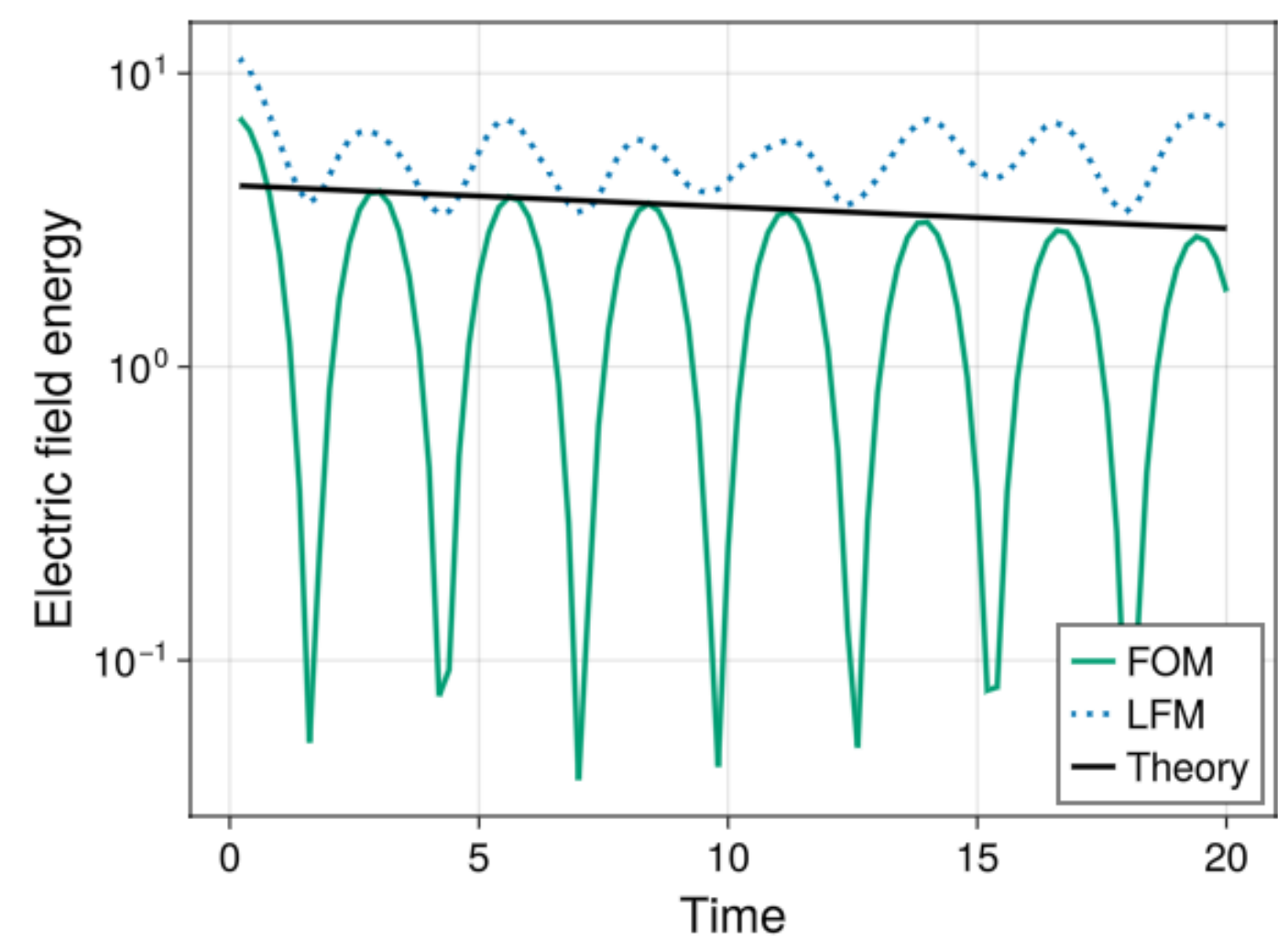


Figure 1. Electric field energy of the full-order model (FOM) using 512,000 particles vs. low-fidelity model (LFM) using 5,120 particles.

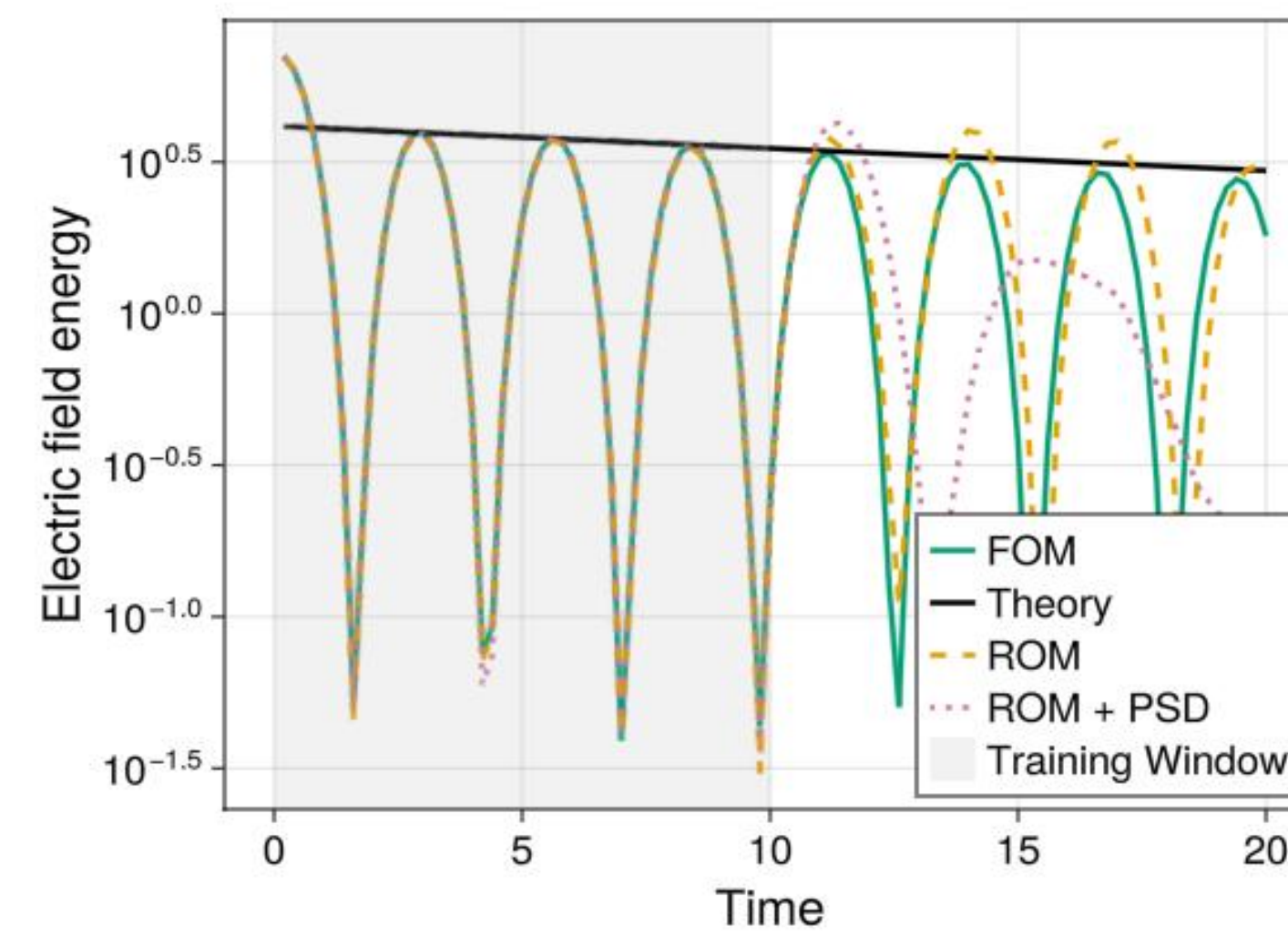


Figure 2. Electric field energy of the FOM using 512,000 particles vs. both ROM variants using 5,000 particles.

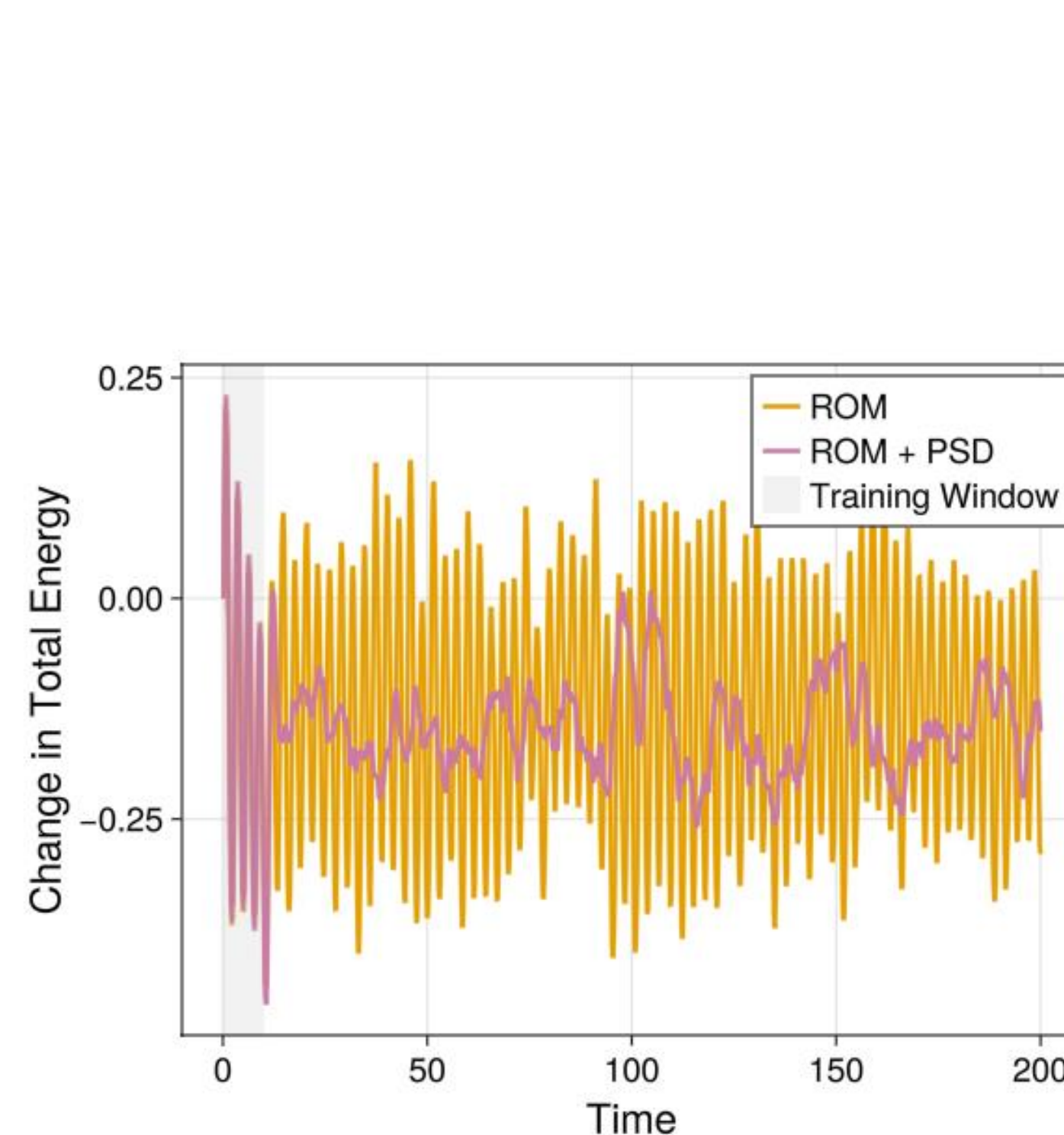


Figure 3. Energy conservation of both ROM variants over a long simulation. *Symplectic structure preservation guarantees bounded energy errors

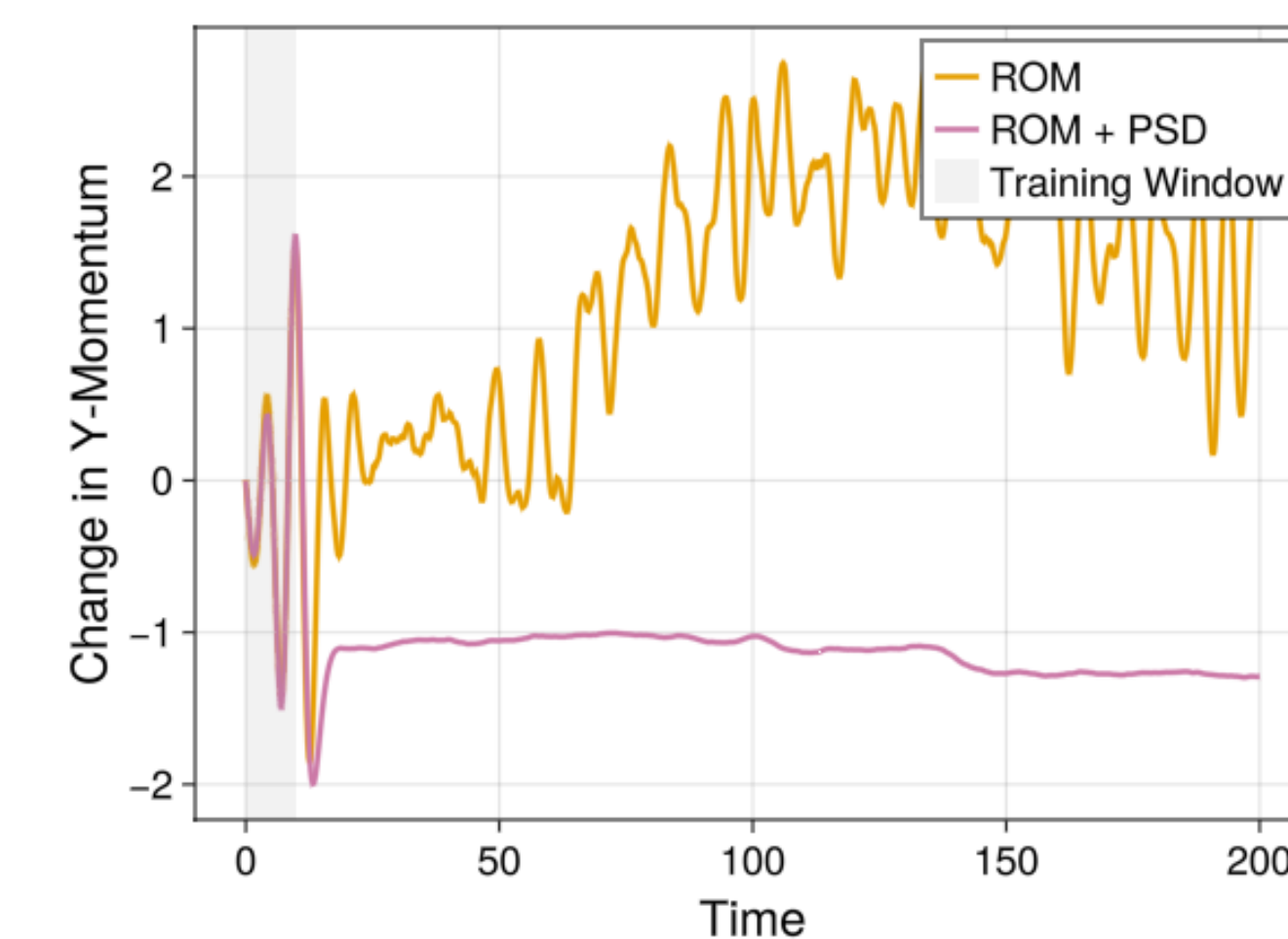
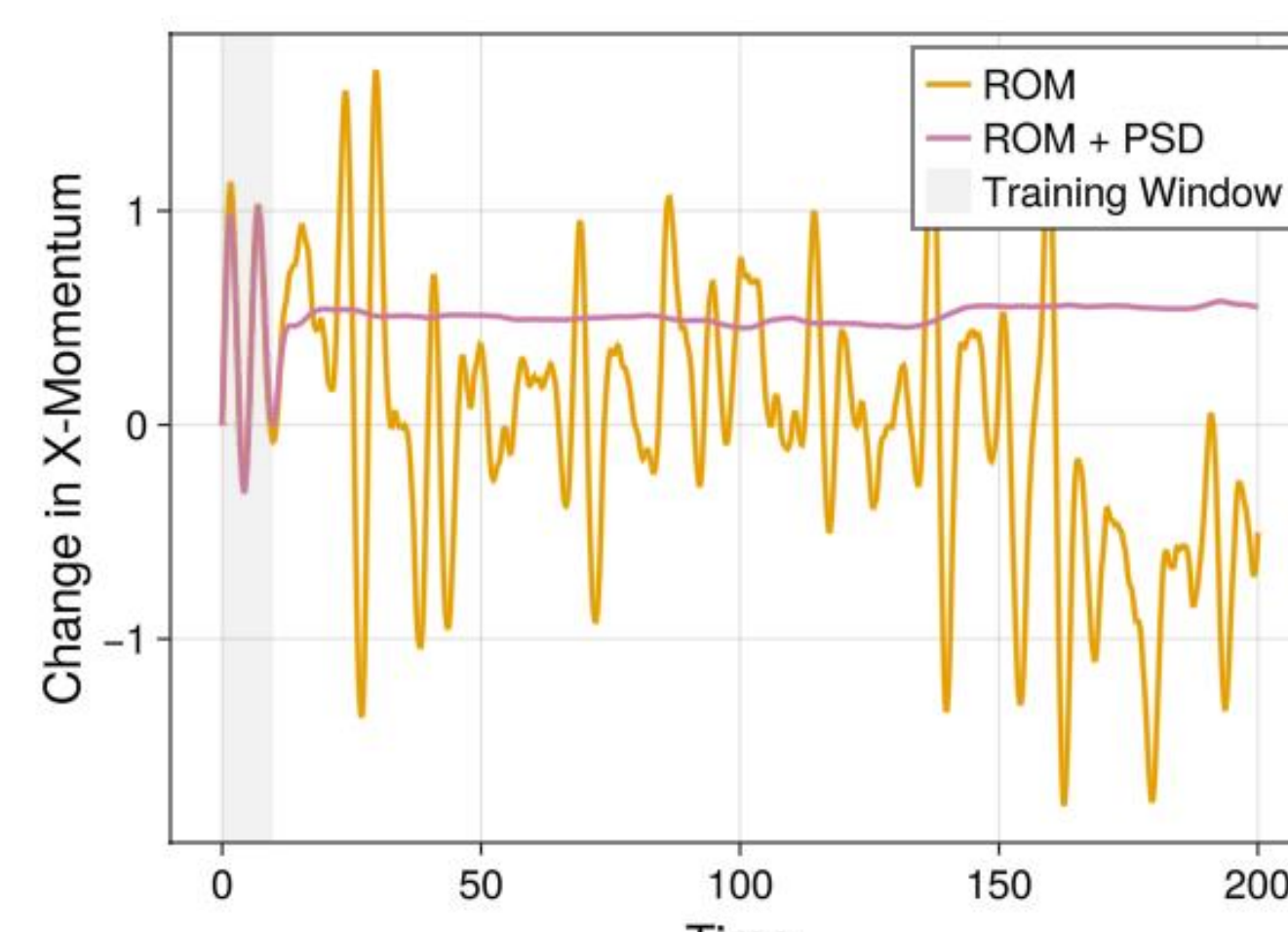


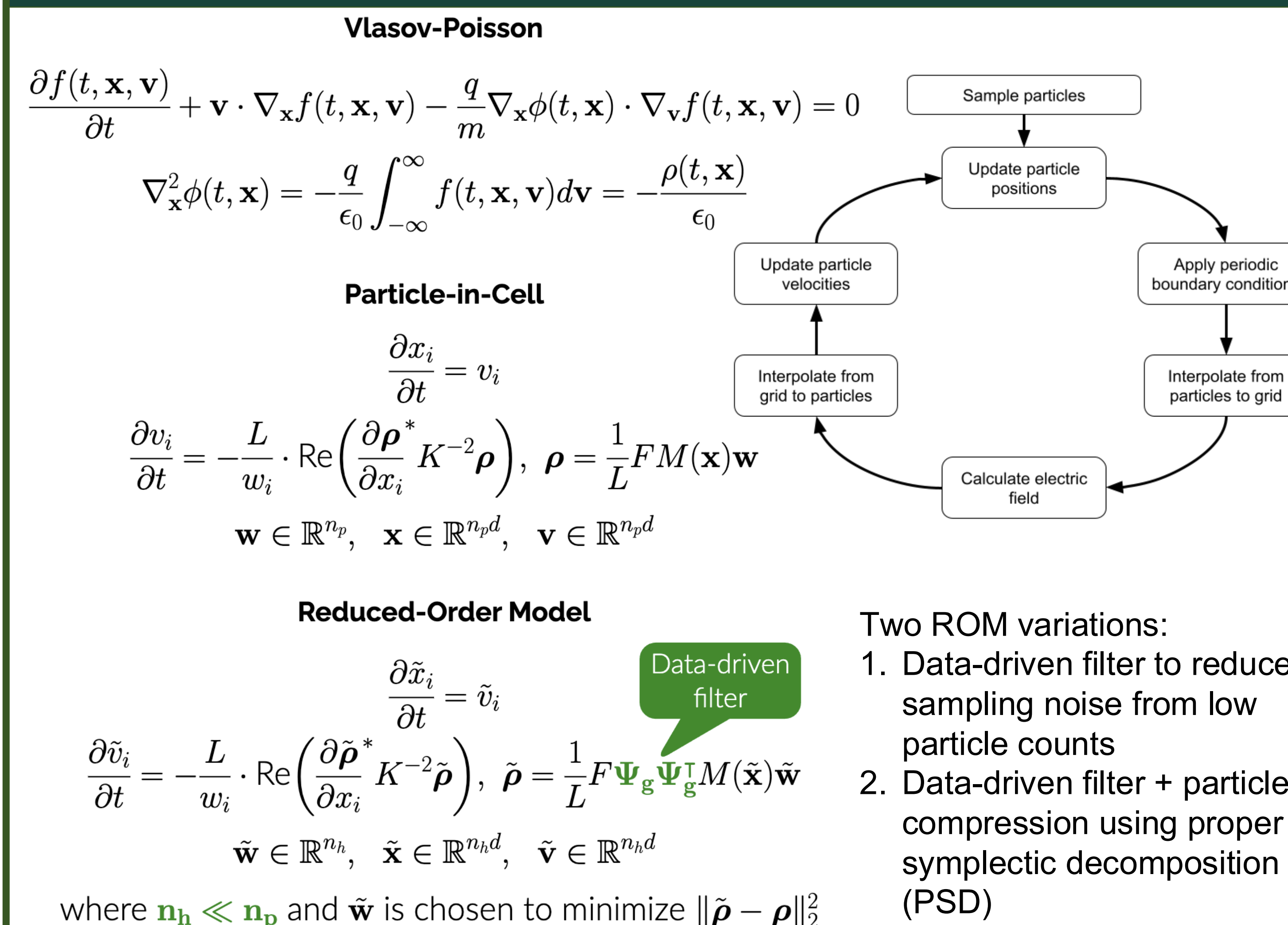
Figure 4. Momentum conservation of both ROM variants over a long simulation.
Top: x-momentum conservation
Bottom: y-momentum conservation
*Momentum conservation is not guaranteed

Motivation

- Fluid models of plasmas are fundamentally limited to capturing macroscopic physics
- Kinetic models capture high-fidelity physics but are often prohibitively expensive
- Structure-preserving reduced-order models (ROMs) potentially address this limitation.

Structure-preserving reduced-order-models yield over 100x speedups with strong guarantees of long-time energy conservation.

Governing Equations



Two-Stream Instability

The two-stream instability is another classic test problem for PIC codes where two counter-streaming beams of particles interact electromagnetically, forming a phase-space vortex. The behavior can only be captured using kinetic plasma physics simulations. Simulated in 2D2V with the initial condition

$$f(t=0, x, y, v_x, v_y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{v_y^2}{2}\right) \left[\exp\left(-\frac{(v_x - v_0)^2}{2}\right) + \exp\left(-\frac{(v_x + v_0)^2}{2}\right) \right] [1 + \alpha \cos(kx)]$$

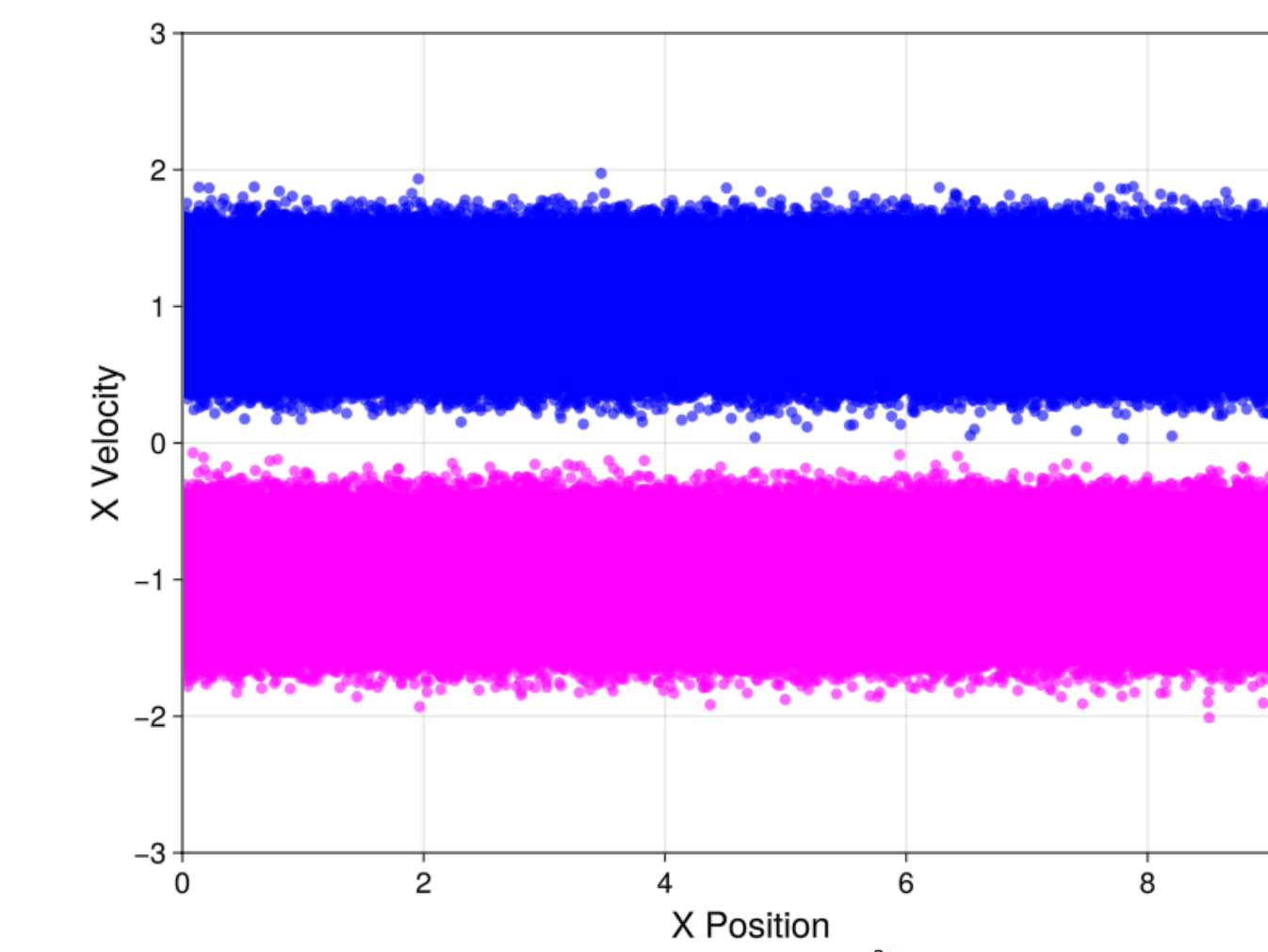


Figure 5. The initial condition for the two-stream instability.

Figure 6. Particle distribution at $t = 20$.
Top-left: FOM with 2,048,000 particles
Top-right: LFM with 8,192 particles
Bottom-left: ROM with 8,000 particles + PSD particle compression
Bottom-right: ROM with 8,000 particles

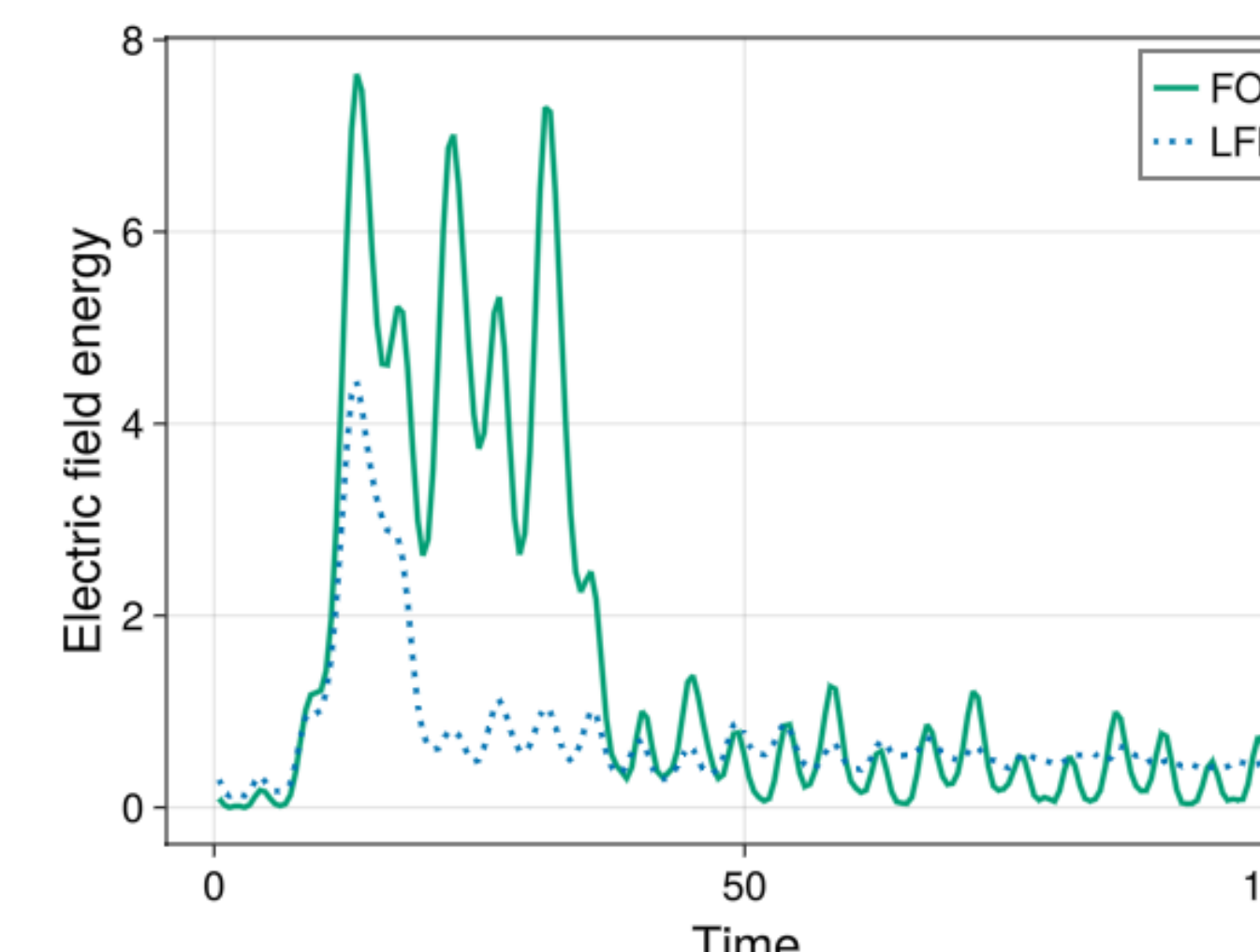
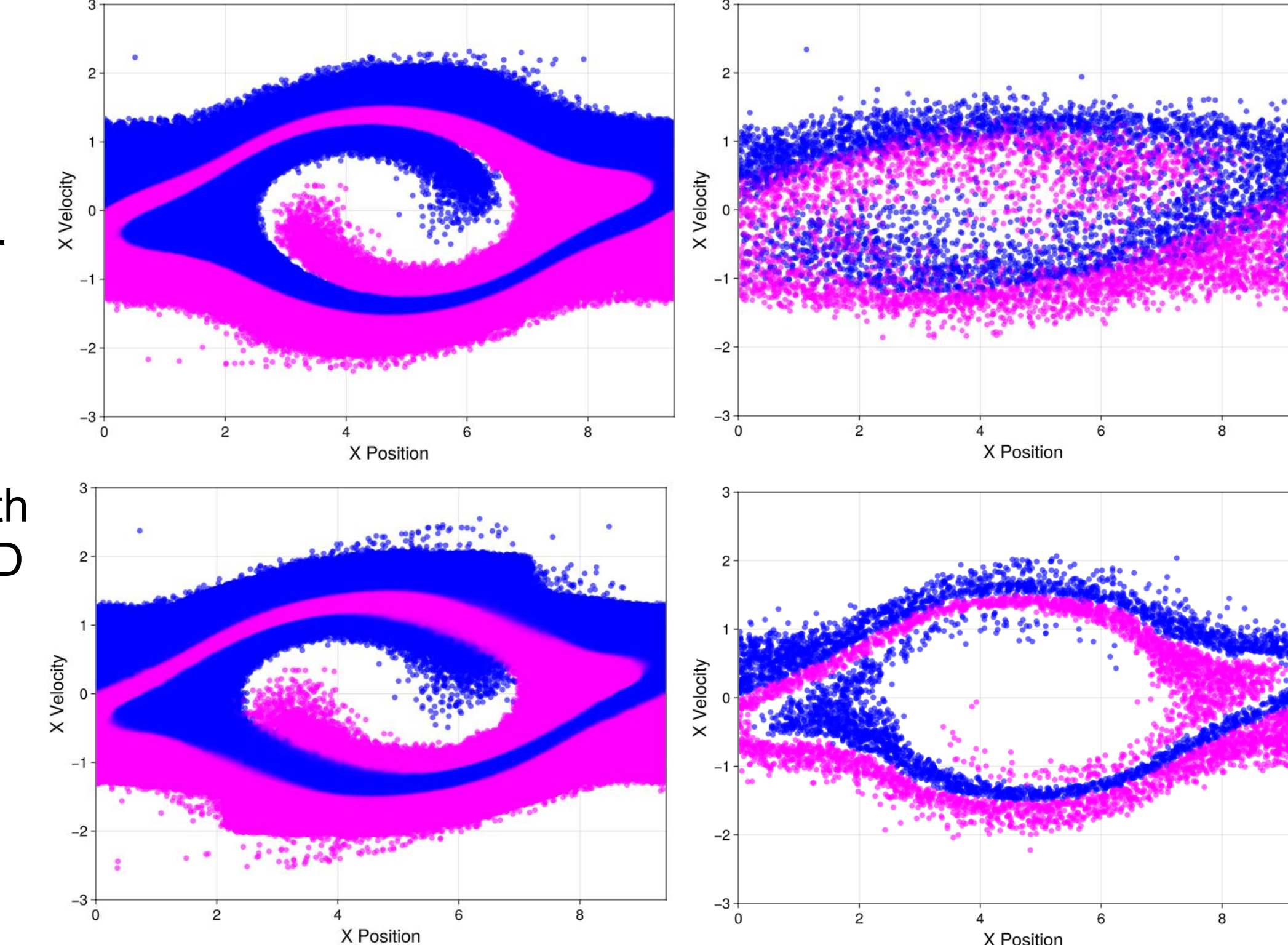


Figure 7. The electric field energy of the FOM using 2,048,000 particles compared with the low-fidelity model (LFM) using 8,192 particles.

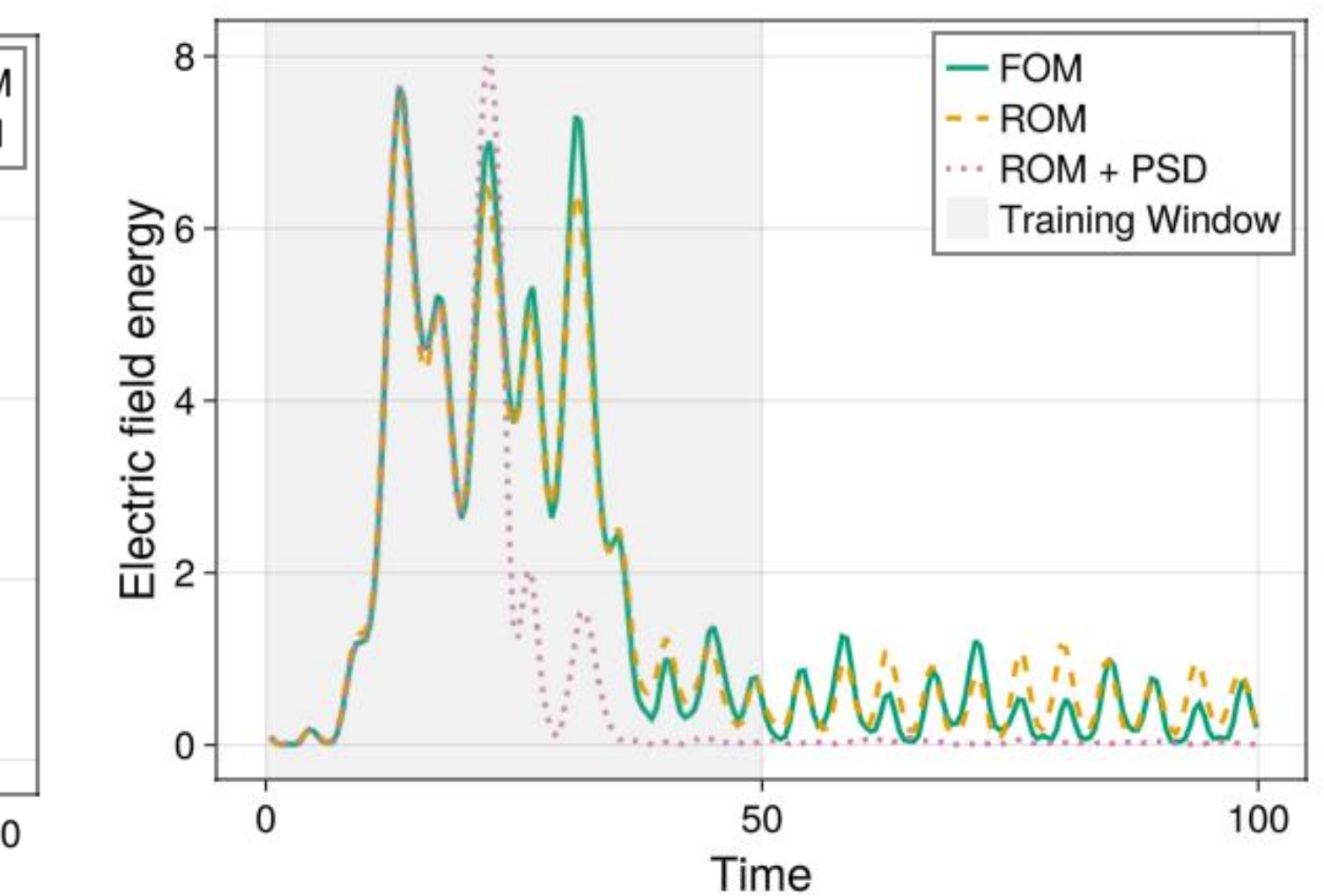


Figure 8. The electric field energy of the FOM using 2,048,000 particles compared with the ROMs using 8,000 particles.